Supplementary material

S1. Validation before the full runs

Each check was run before any full run, and the full runs used only the settings it fixed (Table S1).

S2. Run record

Claude Haiku 4.5 declined to answer for part of the gray-image prompts. Declined answers counted as wrong, the abstention rate is reported for each condition, and the error correlation on answered items is reported as a sensitivity analysis. The output-quality stop that ends a run when more than 15% of answers cannot be parsed was designed to catch configuration errors and was turned off for the gray-image conditions. Each perceiver wrote its findings reports once for each image and slot before any reasoner read a report (12,906 reports; median length 75 to 90 words by perceiver, maximum 118 to 127). After each model finished, every condition was run again to fill episodes lost to transient errors; calls already made were read from the cache.

S3. Detailed results

S4. Open-weight models

S5. Text-only clinical questions

The same reading-panel designs were applied to text-only clinical questions in a companion analysis: 8 language models from OpenAI, Google, Anthropic and DeepSeek answered items from MedQA, MedAgentsBench and the text set of MedXpertQA as single readers and as independent, centralized, discussion and tiered panels of 3 to 9 readers with specialty roles. Over 340 panel configurations, the first-round error correlation between panel positions was 0.758, which leaves 1.27 effective independent readers in a panel

Table S1: Validation checks and the settings they fixed.

Check	Result	Setting fixed
V1	All four API models parsed 100% of single-reader answers through OpenRouter. GPT-5 mini used 256 reasoning tokens at low effort and 2,304 at high effort on the same prompt.	Claude Haiku 4.5 retained; GPT-5 mini at low effort.
V2	Single-reader accuracy relative to the original image, over 10 model-by-data-set cells: CLAHE −0.0 percentage points on average (range −6 to +14), grayscale inversion −6.6 (−12 to +2), 256 pixels −1.0, 128 pixels −5.6.	Inversion exceeded the 5-point limit and was replaced by gamma 0.7: readers 4–6 of the rendering panel saw gamma 0.7, readers 7–9 CLAHE.
V3	With four images per prompt (the query and three references), parsing was 100%. A CLIP cosine threshold of 0.95 removed about 32 ordinary similar cases per query, because radiographs of different patients reach cosines of 0.91–0.95; a perceptual hash alone missed the same CT level and the same patient's chest radiograph (cosine 0.99).	References excluded if SHA-256 identical, within Hamming distance 8 of the 64-bit perceptual hash, or at CLIP cosine of at least 0.98.
V4	SLAKE-target: 335 eligible questions, 250 sampled; in 20 inspected cases the target box covered the queried organ or lesion and the control box did not overlap it and lay within the body. SLAKE-pairs: 200 pairs over 104 question templates.	Used as built.
V5	Findings reports had a median of 72–77 words (maximum 109); answer parsing 100%.	Used as built.
V6	With option-letter readout, the LLaVA family put 96–100% of the probability on "A" regardless of the image. Pooled accuracy on 100 items (50 each from VQA-RAD and SLAKE): MedGemma 84%, HuatuoGPT-Vision 79%, Qwen3-VL 78%, Qwen2-VL 74%, LLaVA-1.5 55%, LLaVA-Med 53%.	Yes/no items read from the "yes"/"no" tokens; LLaVA-Med excluded by the 55% rule.
Throughput	Qwen 3.8 Flash used 250–6,400 reasoning tokens and 6–77 s per call.	Qwen 3.8 Flash ran the original-image, graded-evidence, generic-panel and rendering conditions only.

Table S2: Reading panels with the original images. Accuracy with item bootstrap 95% CI; pp, percentage points; round 0, the independent answers before discussion.

Set	Model	n	Single (%)	Mean panel reader (%)	Panel 9 (%)
VQA-RAD	GPT-5 mini	250	72.0 [66.4, 77.2]	71.3 [66.3, 76.2]	69.2 [63.2, 74.8]
VQA-RAD	Gemini 3.7 Flash	250	81.2 [76.4, 86.0]	81.3 [76.9, 85.4]	82.4 [77.2, 86.8]
VQA-RAD	Claude Haiku 4.5	250	60.8 [54.4, 66.8]	65.9 [61.0, 71.0]	64.8 [58.8, 70.8]
VQA-RAD	Qwen 3.8 Flash	250	69.6 [64.0, 75.2]	79.9 [76.3, 83.3]	83.2 [78.4, 87.6]
SLAKE	GPT-5 mini	250	80.8 [76.0, 85.6]	79.8 [75.4, 84.3]	78.4 [73.2, 83.6]
SLAKE	Gemini 3.7 Flash	250	91.2 [87.6, 94.4]	89.9 [86.7, 93.1]	90.0 [86.4, 93.6]
SLAKE	Claude Haiku 4.5	250	72.4 [67.2, 78.0]	72.3 [67.4, 77.0]	72.4 [66.8, 78.0]
SLAKE	Qwen 3.8 Flash	250	76.4 [70.8, 81.2]	83.3 [79.9, 87.0]	84.4 [80.0, 89.2]
MedXpertQA-MM	GPT-5 mini	250	54.8 [48.8, 60.4]	53.2 [48.1, 58.4]	55.6 [49.6, 61.6]
MedXpertQA-MM	Gemini 3.7 Flash	250	84.4 [79.6, 88.8]	83.3 [79.2, 87.2]	83.6 [78.8, 88.4]
MedXpertQA-MM	Claude Haiku 4.5	250	33.2 [27.6, 38.8]	30.8 [25.4, 36.1]	30.4 [24.4, 36.0]
MedXpertQA-MM	Qwen 3.8 Flash	250	43.6 [37.6, 49.6]	46.7 [41.9, 51.8]	49.2 [43.2, 55.2]

Table S3: Reading panels with the original images (continued).

Set	Model	Discussion 9 (%)	Panel 9 − single (pp)	Mean panel reader − single (pp)	Panel 9 − its mean reader (pp)
VQA-RAD	GPT-5 mini	70.0 [64.0, 75.6]	−2.8 [−6.4, +0.8]	−0.7 [−3.5, +2.2]	−2.1 [−4.0, −0.4]
VQA-RAD	Gemini 3.7 Flash	81.2 [76.0, 86.0]	+1.2 [−2.4, +4.8]	+0.1 [−2.7, +2.7]	+1.1 [−0.5, +2.7]
VQA-RAD	Claude Haiku 4.5	66.4 [60.4, 72.4]	+4.0 [−0.8, +8.8]	+5.1 [+1.0, +9.1]	−1.1 [−3.1, +0.8]
VQA-RAD	Qwen 3.8 Flash	80.4 [75.6, 85.2]	+13.6 [+7.6, +19.6]	+10.3 [+5.2, +15.5]	+3.3 [+0.8, +5.8]
SLAKE	GPT-5 mini	80.8 [75.6, 85.6]	−2.4 [−5.2, +0.4]	−1.0 [−3.5, +1.3]	−1.4 [−2.9, +0.0]
SLAKE	Gemini 3.7 Flash	89.2 [85.2, 93.2]	−1.2 [−3.2, +0.4]	−1.3 [−2.8, +0.1]	+0.1 [−1.0, +1.3]
SLAKE	Claude Haiku 4.5	74.4 [69.2, 79.6]	+0.0 [−4.4, +4.4]	−0.1 [−3.8, +3.6]	+0.1 [−1.6, +1.7]
SLAKE	Qwen 3.8 Flash	90.4 [86.8, 94.0]	+8.0 [+4.0, +12.4]	+6.9 [+3.5, +10.6]	+1.1 [−0.8, +2.9]
MedXpertQA-MM	GPT-5 mini	58.4 [52.4, 64.4]	+0.8 [−3.6, +5.6]	−1.6 [−5.2, +2.4]	+2.4 [+0.1, +4.6]
MedXpertQA-MM	Gemini 3.7 Flash	82.4 [77.6, 87.2]	−0.8 [−4.0, +2.4]	−1.1 [−3.5, +1.2]	+0.3 [−1.4, +2.0]
MedXpertQA-MM	Claude Haiku 4.5	30.0 [24.4, 36.0]	−2.8 [−6.0, +0.4]	−2.4 [−5.4, +0.5]	−0.4 [−1.9, +1.1]
MedXpertQA-MM	Qwen 3.8 Flash	52.4 [45.6, 58.8]	+5.6 [+0.4, +11.2]	+3.1 [−1.5, +7.6]	+2.5 [+0.0, +4.9]

Table S4: Reading panels with the original images (continued).

Set	Model	Discussion 9 − single (pp)	ϕ round 0	N_{eff}	Unanimous r0 (%)
VQA-RAD	GPT-5 mini	−2.0 [−5.6, +1.6]	0.752 [0.697, 0.801]	1.28	74.4
VQA-RAD	Gemini 3.7 Flash	+0.0 [−3.6, +3.6]	0.725 [0.653, 0.785]	1.32	76.0
VQA-RAD	Claude Haiku 4.5	+5.6 [+0.8, +10.8]	0.731 [0.678, 0.781]	1.31	68.0
VQA-RAD	Qwen 3.8 Flash	+10.8 [+5.2, +16.4]	0.439 [0.369, 0.501]	1.99	54.8
SLAKE	GPT-5 mini	+0.0 [−2.8, +2.8]	0.785 [0.729, 0.844]	1.24	82.0
SLAKE	Gemini 3.7 Flash	−2.0 [−4.4, +0.0]	0.785 [0.696, 0.859]	1.24	90.4
SLAKE	Claude Haiku 4.5	+2.0 [−2.0, +6.0]	0.775 [0.723, 0.821]	1.25	75.6
SLAKE	Qwen 3.8 Flash	+14.0 [+9.2, +18.8]	0.529 [0.448, 0.597]	1.72	65.6
MedXpertQA-MM	GPT-5 mini	+3.6 [−0.8, +8.0]	0.685 [0.636, 0.735]	1.39	52.4
MedXpertQA-MM	Gemini 3.7 Flash	−2.0 [−5.2, +0.8]	0.749 [0.669, 0.817]	1.29	82.4
MedXpertQA-MM	Claude Haiku 4.5	−3.2 [−6.8, +0.4]	0.836 [0.790, 0.882]	1.17	71.6
MedXpertQA-MM	Qwen 3.8 Flash	+8.8 [+3.6, +14.4]	0.647 [0.591, 0.701]	1.46	41.2

Table S5: Reading panels with the original images (continued).

Set	Model	Wrong if unanimous r0 (%)	ϕ final	Unanimous final (%)	Wrong if unanimous final (%)
VQA-RAD	GPT-5 mini	20.4	1.000 [1.000, 1.000]	100.0	30.0
VQA-RAD	Gemini 3.7 Flash	11.6	0.997 [0.991, 1.000]	99.6	18.9
VQA-RAD	Claude Haiku 4.5	25.9	0.996 [0.990, 1.000]	99.2	33.5
VQA-RAD	Qwen 3.8 Flash	5.8	0.822 [0.762, 0.874]	86.0	15.8
SLAKE	GPT-5 mini	12.2	0.995 [0.984, 1.000]	99.6	18.9
SLAKE	Gemini 3.7 Flash	5.8	1.000 [1.000, 1.000]	100.0	10.8
SLAKE	Claude Haiku 4.5	22.2	1.000 [1.000, 1.000]	100.0	25.6
SLAKE	Qwen 3.8 Flash	4.3	0.884 [0.811, 0.940]	94.4	8.9
MedXpertQA-MM	GPT-5 mini	35.9	0.996 [0.991, 1.000]	98.8	41.7
MedXpertQA-MM	Gemini 3.7 Flash	9.2	1.000 [1.000, 1.000]	100.0	17.6
MedXpertQA-MM	Claude Haiku 4.5	68.7	1.000 [1.000, 1.000]	100.0	70.0
MedXpertQA-MM	Qwen 3.8 Flash	36.9	0.913 [0.883, 0.941]	82.4	42.7

Table S6: Graded image evidence: single-reader accuracy (%) and error correlation ϕ of the independent panel of 9 at each evidence level; paired differences between the original and the gray image.

Set	Model	Single ≤1024	Single 256	Single 128	Single gray
VQA-RAD	GPT-5 mini	72.0 [66.4, 77.2]	74.0 [68.8, 79.2]	72.8 [67.2, 78.0]	60.8 [54.8, 66.4]
VQA-RAD	Gemini 3.7 Flash	81.2 [76.4, 86.0]	83.6 [78.8, 88.0]	77.6 [72.4, 82.4]	54.4 [48.4, 60.4]
VQA-RAD	Claude Haiku 4.5	60.8 [54.4, 66.8]	63.2 [57.2, 69.2]	65.2 [59.2, 70.8]	46.8 [40.8, 52.8]
VQA-RAD	Qwen 3.8 Flash	69.6 [64.0, 75.2]	67.6 [61.6, 73.2]	72.0 [66.4, 77.6]	54.8 [48.4, 60.8]
SLAKE	GPT-5 mini	80.8 [76.0, 85.6]	78.8 [73.6, 83.6]	75.2 [70.0, 80.4]	54.4 [48.4, 60.8]
SLAKE	Gemini 3.7 Flash	91.2 [87.6, 94.4]	91.2 [87.6, 94.8]	85.2 [80.8, 89.2]	51.2 [44.8, 57.6]
SLAKE	Claude Haiku 4.5	72.4 [67.2, 78.0]	73.6 [68.0, 78.8]	70.0 [64.4, 75.6]	38.8 [32.8, 44.8]
SLAKE	Qwen 3.8 Flash	76.4 [70.8, 81.2]	74.8 [69.2, 80.8]	79.2 [74.4, 84.0]	51.2 [44.8, 57.2]
MedXpertQA-MM	GPT-5 mini	54.8 [48.8, 60.4]	50.4 [44.4, 56.8]	50.8 [44.8, 56.8]	50.8 [44.8, 56.4]
MedXpertQA-MM	Gemini 3.7 Flash	84.4 [79.6, 88.8]	80.4 [75.2, 85.2]	78.4 [73.2, 83.2]	66.4 [60.8, 71.6]
MedXpertQA-MM	Claude Haiku 4.5	33.2 [27.6, 38.8]	32.0 [26.4, 37.6]	32.0 [26.4, 37.6]	29.6 [24.0, 35.2]
MedXpertQA-MM	Qwen 3.8 Flash	43.6 [37.6, 49.6]	40.8 [34.8, 47.2]	43.2 [37.2, 49.6]	41.6 [35.6, 47.6]

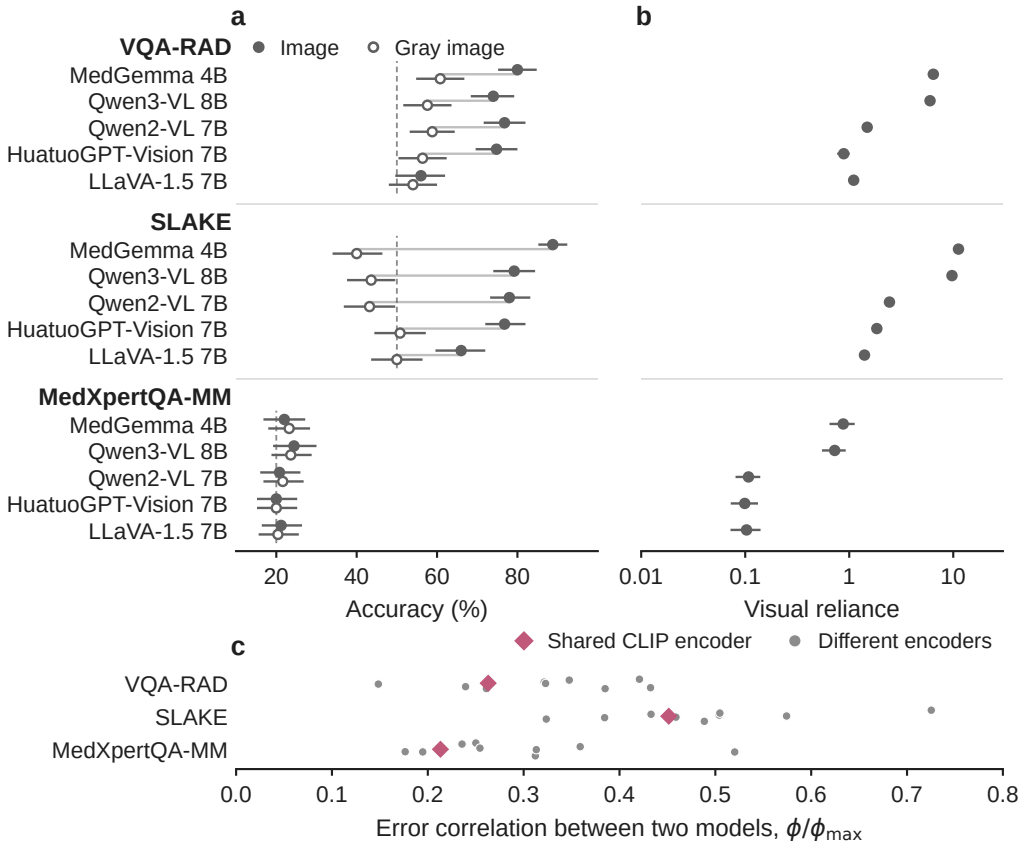


Figure S1: Open-weight models, one forward pass per item. (a) Accuracy with the image (filled) and with a uniform gray image (open); dashed lines, chance. (b) Visual reliance: absolute change in the log-odds of "yes" between the image and the gray image on yes/no items, Kullback–Leibler divergence on MedXpertQA-MM; mean with item bootstrap 95% CI, log scale. (c) Error correlation between two models divided by its maximum for their error rates ($\phi/\phi_{\max}$); diamonds, the pair that shares the CLIP ViT-L/14-336 vision encoder.

Table S7: Graded image evidence (continued).

Set	Model	$\phi \leq 1024$	ϕ 256	ϕ 128	ϕ gray
VQA-RAD	GPT-5 mini	0.752 [0.697, 0.801]	0.694 [0.636, 0.751]	0.644 [0.586, 0.703]	0.742 [0.689, 0.789]
VQA-RAD	Gemini 3.7 Flash	0.725 [0.653, 0.785]	0.726 [0.646, 0.793]	0.749 [0.698, 0.804]	0.922 [0.889, 0.951]
VQA-RAD	Claude Haiku 4.5	0.731 [0.678, 0.781]	0.706 [0.651, 0.756]	0.591 [0.539, 0.650]	0.876 [0.840, 0.911]
VQA-RAD	Qwen 3.8 Flash	0.439 [0.369, 0.501]	0.442 [0.377, 0.501]	0.480 [0.415, 0.541]	0.841 [0.805, 0.879]
SLAKE	GPT-5 mini	0.785 [0.729, 0.844]	0.792 [0.741, 0.845]	0.709 [0.645, 0.761]	0.802 [0.755, 0.845]
SLAKE	Gemini 3.7 Flash	0.785 [0.696, 0.859]	0.803 [0.716, 0.869]	0.820 [0.753, 0.872]	0.982 [0.968, 0.994]
SLAKE	Claude Haiku 4.5	0.775 [0.723, 0.821]	0.752 [0.695, 0.811]	0.718 [0.664, 0.768]	0.879 [0.841, 0.915]
SLAKE	Qwen 3.8 Flash	0.529 [0.448, 0.597]	0.547 [0.479, 0.607]	0.490 [0.408, 0.560]	0.831 [0.788, 0.866]
MedXpertQA-MM	GPT-5 mini	0.685 [0.636, 0.735]	0.690 [0.637, 0.739]	0.700 [0.652, 0.752]	0.730 [0.679, 0.780]
MedXpertQA-MM	Gemini 3.7 Flash	0.749 [0.669, 0.817]	0.762 [0.698, 0.823]	0.784 [0.715, 0.841]	0.799 [0.748, 0.844]
MedXpertQA-MM	Claude Haiku 4.5	0.836 [0.790, 0.882]	0.821 [0.776, 0.868]	0.840 [0.795, 0.881]	0.818 [0.765, 0.861]
MedXpertQA-MM	Qwen 3.8 Flash	0.647 [0.591, 0.701]	0.630 [0.573, 0.688]	0.609 [0.556, 0.667]	0.656 [0.602, 0.709]

of 9. Over the 96 discussion configurations, discussion raised the error correlation from 0.758 (SD 0.117) to 0.990 (SD 0.010), or 1.01 effective readers. Error correlation and its rise during discussion therefore had the same magnitude without images as in the imaging experiments.

Table S8: Graded image evidence (continued).

Set	Model	Single real − gray (pp)	Panel 9 real − gray (pp)	Abstention gray (%)	ϕ gray, answered only
VQA-RAD	GPT-5 mini	+11.2 [+4.0, +18.4]	+7.6 [+0.4, +14.8]	–	–
VQA-RAD	Gemini 3.7 Flash	+26.8 [+18.8, +35.2]	+28.4 [+20.8, +36.8]	–	–
VQA-RAD	Claude Haiku 4.5	+14.0 [+6.0, +22.0]	+10.4 [+2.8, +18.4]	10.4	0.963 [0.942, 0.982]
VQA-RAD	Qwen 3.8 Flash	+14.8 [+6.4, +23.2]	+26.0 [+18.4, +34.0]	0.1	–
SLAKE	GPT-5 mini	+26.4 [+18.8, +33.2]	+25.6 [+18.0, +33.2]	–	–
SLAKE	Gemini 3.7 Flash	+40.0 [+33.2, +46.8]	+39.2 [+32.4, +46.0]	–	–
SLAKE	Claude Haiku 4.5	+33.6 [+24.0, +42.8]	+28.4 [+19.6, +37.2]	4.3	0.931 [0.901, 0.956]
SLAKE	Qwen 3.8 Flash	+25.2 [+16.4, +33.6]	+33.6 [+25.6, +42.0]	–	–
MedXpertQA-MM	GPT-5 mini	+4.0 [−2.0, +10.0]	+6.8 [+1.6, +12.0]	–	–
MedXpertQA-MM	Gemini 3.7 Flash	+18.0 [+12.0, +23.6]	+17.6 [+12.0, +23.6]	–	–
MedXpertQA-MM	Claude Haiku 4.5	+3.6 [−0.4, +8.0]	+0.0 [−4.4, +4.4]	14.8	0.852 [0.803, 0.891]
MedXpertQA-MM	Qwen 3.8 Flash	+2.0 [−4.4, +8.0]	+9.6 [+4.8, +14.8]	0.5	–

Table S9: Reader answers of the independent panel of 9 cross-tabulated by correctness with the original and with the gray image (% of answers), and consensus errors (at least 7 of 9 readers wrong) whose majority answer equals the gray-image majority.

Set	Model	Right with and without image	Right only with image	Wrong only with image	Wrong with and without image
VQA-RAD	GPT-5 mini	47.2	24.1	12.0	16.7
VQA-RAD	Gemini 3.7 Flash	44.0	37.3	10.4	8.3
VQA-RAD	Claude Haiku 4.5	36.6	29.3	13.6	20.5
VQA-RAD	Qwen 3.8 Flash	46.0	33.9	10.8	9.3
SLAKE	GPT-5 mini	43.5	36.3	9.5	10.7
SLAKE	Gemini 3.7 Flash	46.7	43.2	4.4	5.8
SLAKE	Claude Haiku 4.5	25.9	46.4	15.8	11.9
SLAKE	Qwen 3.8 Flash	38.5	44.8	12.0	4.7
MedXpertQA-MM	GPT-5 mini	39.7	13.5	7.9	38.9
MedXpertQA-MM	Gemini 3.7 Flash	61.0	22.3	4.4	12.3
MedXpertQA-MM	Claude Haiku 4.5	22.7	8.1	5.2	64.0
MedXpertQA-MM	Qwen 3.8 Flash	32.9	13.8	7.0	46.4

Table S10: Reader answers of the independent panel of 9 cross-tabulated by correctness with the original and with the gray image (% of answers), and consensus errors (at least 7 of 9 readers wrong) whose majority answer equals the gray-image majority (continued).

Set	Model	Consensus errors (n)	Consensus errors = gray-image majority (%)
VQA-RAD	GPT-5 mini	57	54.4 [42.1, 66.7]
VQA-RAD	Gemini 3.7 Flash	35	42.9 [25.7, 60.0]
VQA-RAD	Claude Haiku 4.5	70	54.3 [42.9, 65.7]
VQA-RAD	Qwen 3.8 Flash	18	44.4 [22.2, 66.7]
SLAKE	GPT-5 mini	42	57.1 [42.9, 71.4]
SLAKE	Gemini 3.7 Flash	19	57.9 [36.8, 78.9]
SLAKE	Claude Haiku 4.5	56	33.9 [21.4, 46.4]
SLAKE	Qwen 3.8 Flash	21	14.3 [0.0, 28.6]
MedXpertQA-MM	GPT-5 mini	95	64.2 [54.7, 72.7]
MedXpertQA-MM	Gemini 3.7 Flash	31	74.2 [58.1, 90.3]
MedXpertQA-MM	Claude Haiku 4.5	163	71.2 [64.4, 77.9]
MedXpertQA-MM	Qwen 3.8 Flash	105	76.2 [67.6, 83.8]

Table S11: Region occlusion on SLAKE-target: share of answers changed by covering the target box or an equal-size control box, and region dependence (target minus control).

Model	Arm	Target flip (%)	Control flip (%)	Region dependence (pp)	Single target flip (%)
GPT-5 mini	API, panel readers	23.5 [19.7, 27.6]	6.4 [4.7, 8.4]	+17.1 [+13.2, +21.2]	26.4 [20.8, 32.0]
Gemini 3.7 Flash	API, panel readers	36.4 [31.6, 41.4]	3.6 [2.4, 4.9]	+32.9 [+28.2, +38.0]	36.8 [30.8, 42.8]
Claude Haiku 4.5	API, panel readers	10.7 [8.2, 13.6]	7.5 [5.5, 9.6]	+3.2 [+0.8, +5.7]	8.4 [5.2, 12.0]
HuatuoGPT-Vision 7B	open weight	28.0 [22.4, 34.0]	6.4 [3.6, 9.6]	+21.6 [+15.6, +28.0]	
LLaVA-1.5 7B	open weight	12.8 [8.8, 17.2]	10.4 [6.8, 14.4]	+2.4 [−2.0, +6.8]	
MedGemma 4B	open weight	13.2 [9.2, 17.6]	1.6 [0.4, 3.2]	+11.6 [+7.6, +15.6]	
Qwen2-VL 7B	open weight	11.2 [7.6, 15.2]	4.8 [2.4, 7.6]	+6.4 [+2.0, +10.4]	
Qwen3-VL 8B	open weight	23.6 [18.4, 28.8]	3.6 [1.6, 6.0]	+20.0 [+14.4, +26.0]	

Table S12: Region occlusion on SLAKE-target (continued).

Model	Arm	Single control flip (%)	Single region dependence (pp)
GPT-5 mini	API, panel readers	8.4 [5.2, 12.0]	+18.0 [+11.6, +24.8]
Gemini 3.7 Flash	API, panel readers	2.8 [1.2, 4.8]	+34.0 [+28.0, +40.4]
Claude Haiku 4.5	API, panel readers	6.4 [3.6, 9.2]	+2.0 [−1.6, +5.2]
HuatuoGPT-Vision 7B	open weight		
LLaVA-1.5 7B	open weight		
MedGemma 4B	open weight		
Qwen2-VL 7B	open weight		
Qwen3-VL 8B	open weight		

Table S13: Same question, different image (SLAKE-pairs, 200 pairs with opposite reference answers).

Model	Single: both correct (%)	Single: answers differ (%)	Panel 9: both correct (%)	Panel 9: answers differ (%)
GPT-5 mini	59.0 [52.0, 65.5]	62.0 [55.5, 68.5]	62.0 [55.5, 68.5]	65.0 [58.0, 71.0]
Gemini 3.7 Flash	74.0 [67.5, 80.0]	75.0 [69.0, 81.0]	74.5 [68.0, 80.5]	75.5 [69.0, 81.5]
Claude Haiku 4.5	38.5 [31.5, 45.0]	42.5 [35.5, 49.5]	42.5 [35.5, 49.0]	46.0 [39.0, 52.5]
HuatuoGPT-Vision 7B (open weight)	48.5 [41.5, 55.5]	53.0 [46.0, 60.0]		
LLaVA-1.5 7B (open weight)	31.5 [25.5, 38.0]	39.5 [33.0, 46.5]		
MedGemma 4B (open weight)	62.5 [56.0, 68.5]	63.5 [57.0, 69.5]		
Qwen2-VL 7B (open weight)	49.5 [42.5, 56.0]	53.5 [46.5, 60.0]		
Qwen3-VL 8B (open weight)	53.5 [46.5, 60.5]	56.5 [49.5, 63.0]		

Table S14: Same question, different image (SLAKE-pairs, 200 pairs with opposite reference answers) (continued).

Model	Readers: answers differ (%)
GPT-5 mini	66.0 [60.4, 71.2]
Gemini 3.7 Flash	75.9 [70.1, 81.4]
Claude Haiku 4.5	45.3 [39.2, 51.3]
HuatuoGPT-Vision 7B (open weight)	
LLaVA-1.5 7B (open weight)	
MedGemma 4B (open weight)	
Qwen2-VL 7B (open weight)	
Qwen3-VL 8B (open weight)	

Table S15: Discussion: readers correct in round 0 who ended the discussion wrong, by whether their round-0 answer differed from their gray-image answer and by round-0 position.

Set	Model	Abandoned, from image (%)	Abandoned, same as prior (%)	Difference (pp)	Abandoned, in minority (%)
VQA-RAD	GPT-5 mini	9.8 [5.5, 14.9]	4.6 [2.3, 7.4]	+5.1 [+0.4, +10.4]	72.3 [55.2, 88.9]
VQA-RAD	Gemini 3.7 Flash	3.7 [1.4, 6.2]	3.4 [1.6, 5.7]	+0.3 [−2.9, +3.3]	100.0 [100.0, 100.0]
VQA-RAD	Claude Haiku 4.5	7.1 [3.9, 11.0]	5.7 [3.1, 8.9]	+1.4 [−2.9, +5.9]	75.0 [59.2, 89.7]
VQA-RAD	Qwen 3.8 Flash	12.5 [7.8, 17.7]	6.6 [3.8, 9.8]	+5.9 [+0.6, +11.5]	54.1 [37.4, 70.6]
SLAKE	GPT-5 mini	4.5 [2.1, 7.3]	1.9 [0.8, 3.3]	+2.6 [+0.3, +5.2]	56.1 [34.9, 78.8]
SLAKE	Gemini 3.7 Flash	2.9 [0.7, 5.3]	1.4 [0.3, 2.9]	+1.5 [−1.1, +4.2]	71.4 [41.2, 100.0]
SLAKE	Claude Haiku 4.5	4.0 [1.9, 6.7]	3.4 [1.1, 6.4]	+0.6 [−3.3, +4.1]	62.5 [41.4, 83.0]
SLAKE	Qwen 3.8 Flash	4.2 [2.0, 6.9]	2.8 [1.3, 4.6]	+1.4 [−1.4, +4.6]	41.5 [23.7, 61.2]
MedXpertQA-MM	GPT-5 mini	18.1 [11.5, 26.1]	2.7 [1.4, 4.3]	+15.4 [+9.6, +23.1]	68.5 [52.6, 85.3]
MedXpertQA-MM	Gemini 3.7 Flash	7.2 [3.7, 11.6]	2.6 [1.2, 4.5]	+4.6 [+0.8, +9.0]	71.7 [48.0, 93.9]
MedXpertQA-MM	Claude Haiku 4.5	23.6 [12.8, 36.3]	6.3 [2.9, 11.1]	+17.4 [+6.4, +29.7]	83.9 [67.3, 100.0]
MedXpertQA-MM	Qwen 3.8 Flash	18.1 [12.3, 25.1]	5.4 [2.9, 8.5]	+12.7 [+6.6, +19.8]	62.2 [48.6, 76.7]

Table S16: Discussion (continued).

Set	Model	Abandoned, in majority (%)	From image: in minority (%)	Same as prior: in minority (%)	n correct (image/prior)
VQA-RAD	GPT-5 mini	1.9 [0.4, 3.8]	11.2 [6.8, 16.4]	3.8 [2.0, 5.9]	543/1062
VQA-RAD	Gemini 3.7 Flash	1.0 [0.0, 2.3]	2.5 [0.8, 4.5]	2.6 [1.1, 4.5]	840/989
VQA-RAD	Claude Haiku 4.5	1.4 [0.0, 3.2]	8.5 [5.2, 12.8]	5.3 [3.0, 8.0]	660/823
VQA-RAD	Qwen 3.8 Flash	6.5 [3.9, 9.4]	6.2 [3.4, 9.4]	4.9 [2.6, 7.7]	762/1035
SLAKE	GPT-5 mini	1.1 [0.0, 2.5]	4.0 [2.1, 6.4]	3.4 [1.7, 5.5]	817/978
SLAKE	Gemini 3.7 Flash	1.2 [0.2, 2.4]	1.8 [0.5, 3.5]	1.0 [0.3, 2.1]	971/1051
SLAKE	Claude Haiku 4.5	1.4 [0.3, 2.9]	3.4 [1.8, 5.7]	4.8 [2.0, 8.3]	1044/583
SLAKE	Qwen 3.8 Flash	1.8 [0.7, 3.1]	3.4 [1.6, 5.5]	5.5 [3.3, 8.5]	1009/866
MedXpertQA-MM	GPT-5 mini	1.4 [0.0, 3.4]	20.1 [13.3, 28.4]	3.5 [1.8, 5.5]	304/894
MedXpertQA-MM	Gemini 3.7 Flash	1.9 [0.6, 3.5]	6.6 [3.3, 11.0]	1.5 [0.6, 2.6]	501/1373
MedXpertQA-MM	Claude Haiku 4.5	4.4 [0.9, 9.1]	19.8 [11.1, 31.0]	3.9 [1.7, 6.9]	182/511
MedXpertQA-MM	Qwen 3.8 Flash	2.4 [0.7, 4.5]	23.5 [16.3, 32.1]	6.2 [3.6, 9.4]	310/740

Table S17: Seeing and judging: mean reader accuracy (%) when 9 readers each read their own question-blind findings report from a perceiver, against reading the image itself; error correlation ϕ.

Set	Reasoner	Reports by GPT-5 mini	Reports by Gemini	Reports by Haiku	Image itself
VQA-RAD	GPT-5 mini	69.0 [63.7, 73.9]	78.7 [74.1, 82.8]	61.6 [56.4, 66.7]	72.6 [67.6, 77.3]
VQA-RAD	Gemini 3.7 Flash	70.2 [64.6, 75.2]	81.3 [76.9, 85.4]	63.6 [58.4, 68.6]	80.3 [75.6, 84.5]
VQA-RAD	Claude Haiku 4.5	70.6 [65.4, 75.6]	78.8 [74.2, 83.1]	63.5 [58.1, 68.4]	62.5 [56.7, 68.0]
SLAKE	GPT-5 mini	83.5 [79.3, 87.5]	87.4 [83.7, 90.9]	78.8 [74.3, 82.9]	79.1 [74.5, 83.6]
SLAKE	Gemini 3.7 Flash	84.9 [81.0, 88.7]	88.2 [84.6, 91.6]	80.2 [76.0, 84.4]	89.7 [86.3, 93.1]
SLAKE	Claude Haiku 4.5	81.0 [76.8, 85.1]	84.3 [80.4, 88.1]	76.3 [71.8, 80.8]	73.4 [68.1, 78.5]
MedXpertQA-MM	GPT-5 mini	50.1 [45.0, 55.2]	62.0 [56.7, 67.2]	45.0 [40.1, 49.9]	52.9 [47.7, 58.0]
MedXpertQA-MM	Gemini 3.7 Flash	60.8 [55.6, 65.9]	74.5 [69.6, 79.1]	57.8 [52.7, 62.7]	83.7 [79.4, 87.8]
MedXpertQA-MM	Claude Haiku 4.5	30.6 [25.6, 35.7]	45.6 [40.0, 50.9]	27.4 [22.6, 32.2]	31.7 [26.3, 37.3]

Table S18: Seeing and judging (continued).

Set	Reasoner	GPT-5 mini reports − image (pp)	Gemini reports − image (pp)	Haiku reports − image (pp)	ϕ image
VQA-RAD	GPT-5 mini	−3.6 [−8.4, +1.4]	+6.1 [+1.4, +10.9]	−10.9 [−16.8, −5.0]	0.741 [0.683, 0.797]
VQA-RAD	Gemini 3.7 Flash	−10.1 [−16.1, −4.0]	+1.0 [−4.0, +6.0]	−16.7 [−23.4, −10.1]	0.776 [0.715, 0.836]
VQA-RAD	Claude Haiku 4.5	+8.1 [+1.4, +14.9]	+16.3 [+9.9, +22.6]	+0.9 [−5.3, +6.8]	0.878 [0.835, 0.911]
SLAKE	GPT-5 mini	+4.4 [+0.9, +8.2]	+8.3 [+4.4, +12.4]	−0.4 [−5.0, +4.6]	0.800 [0.746, 0.844]
SLAKE	Gemini 3.7 Flash	−4.8 [−8.5, −1.3]	−1.6 [−4.0, +1.2]	−9.5 [−13.9, −4.9]	0.828 [0.744, 0.895]
SLAKE	Claude Haiku 4.5	+7.6 [+3.1, +12.1]	+10.9 [+6.4, +15.7]	+2.9 [−1.6, +7.3]	0.882 [0.838, 0.922]
MedXpertQA-MM	GPT-5 mini	−2.8 [−6.9, +1.0]	+9.1 [+3.4, +14.4]	−7.9 [−12.7, −3.2]	0.692 [0.642, 0.742]
MedXpertQA-MM	Gemini 3.7 Flash	−22.9 [−28.4, −17.5]	−9.2 [−13.3, −5.2]	−25.9 [−31.2, −20.4]	0.811 [0.742, 0.866]
MedXpertQA-MM	Claude Haiku 4.5	−1.1 [−5.9, +3.6]	+13.8 [+7.8, +19.5]	−4.4 [−9.0, −0.0]	0.915 [0.876, 0.949]

Table S19: Seeing and judging (continued).

Set	Reasoner	ϕ own reports (same model)	ϕ shared report
VQA-RAD	GPT-5 mini	0.818 [0.770, 0.866]	0.944 [0.917, 0.970]
VQA-RAD	Gemini 3.7 Flash	0.783 [0.727, 0.841]	0.990 [0.975, 1.000]
VQA-RAD	Claude Haiku 4.5	0.723 [0.668, 0.773]	0.986 [0.973, 0.998]
SLAKE	GPT-5 mini	0.777 [0.711, 0.842]	0.955 [0.921, 0.985]
SLAKE	Gemini 3.7 Flash	0.771 [0.687, 0.840]	0.965 [0.934, 0.988]
SLAKE	Claude Haiku 4.5	0.728 [0.665, 0.785]	0.969 [0.943, 0.990]
MedXpertQA-MM	GPT-5 mini	0.641 [0.595, 0.696]	0.800 [0.755, 0.845]
MedXpertQA-MM	Gemini 3.7 Flash	0.715 [0.654, 0.770]	0.880 [0.831, 0.927]
MedXpertQA-MM	Claude Haiku 4.5	0.701 [0.644, 0.758]	0.893 [0.857, 0.933]

Table S20: What two readers share, VQA-RAD: matched within-model contrasts of mean pairwise error correlation ($\Delta\phi$); sampling only, the baseline ϕ of generic-panel readers of one model.

Component	Model or model pair	$\Delta\phi$ (sampling only: ϕ)
sampling only	Claude Haiku 4.5	0.878 [0.836, 0.918]
sampling only	Gemini 3.7 Flash	0.776 [0.714, 0.825]
sampling only	GPT-5 mini	0.741 [0.676, 0.792]
sampling only	Qwen 3.8 Flash	0.442 [0.378, 0.504]
role	Mean over models	−0.048 [−0.071, −0.023]
role	Claude Haiku 4.5	−0.148 [−0.198, −0.098]
role	Gemini 3.7 Flash	−0.051 [−0.104, −0.002]
role	GPT-5 mini	+0.011 [−0.029, +0.053]
role	Qwen 3.8 Flash	−0.003 [−0.049, +0.042]
rendering	Mean over models	−0.092 [−0.123, −0.068]
rendering	Claude Haiku 4.5	−0.151 [−0.211, −0.101]
rendering	Gemini 3.7 Flash	−0.181 [−0.260, −0.125]
rendering	GPT-5 mini	−0.061 [−0.103, −0.014]
rendering	Qwen 3.8 Flash	+0.023 [−0.003, +0.048]
reference cases	Mean over models	−0.110 [−0.142, −0.084]
reference cases	Claude Haiku 4.5	−0.229 [−0.273, −0.191]
reference cases	Gemini 3.7 Flash	−0.043 [−0.093, +0.001]
reference cases	GPT-5 mini	−0.058 [−0.102, −0.023]
report	Mean over models	−0.199 [−0.233, −0.164]
report	Claude Haiku 4.5	−0.263 [−0.314, −0.212]
report	Gemini 3.7 Flash	−0.207 [−0.266, −0.151]
report	GPT-5 mini	−0.126 [−0.170, −0.085]
perceiver	Mean over models	−0.311 [−0.362, −0.260]
perceiver	Claude Haiku 4.5	−0.315 [−0.369, −0.256]
perceiver	Gemini 3.7 Flash	−0.322 [−0.375, −0.264]
perceiver	GPT-5 mini	−0.298 [−0.354, −0.248]
reasoner	Mean over models	−0.108 [−0.132, −0.083]
reasoner	Claude Haiku 4.5 and Gemini 3.7 Flash	−0.132 [−0.167, −0.104]
reasoner	Claude Haiku 4.5 and GPT-5 mini	−0.125 [−0.159, −0.093]
reasoner	Gemini 3.7 Flash and GPT-5 mini	−0.066 [−0.098, −0.039]
model	Mean over models	−0.470 [−0.522, −0.421]
model	Claude Haiku 4.5 and Gemini 3.7 Flash	−0.696

Table S21: What two readers share, SLAKE: matched within-model contrasts of mean pairwise error correlation ($\Delta\phi$); sampling only, the baseline ϕ of generic-panel readers of one model.

Component	Model or model pair	$\Delta\phi$ (sampling only: ϕ)
sampling only	Claude Haiku 4.5	0.882 [0.836, 0.921]
sampling only	Gemini 3.7 Flash	0.828 [0.754, 0.906]
sampling only	GPT-5 mini	0.800 [0.751, 0.847]
sampling only	Qwen 3.8 Flash	0.604 [0.530, 0.666]
role	Mean over models	−0.060 [−0.087, −0.036]
role	Claude Haiku 4.5	−0.107 [−0.149, −0.069]
role	Gemini 3.7 Flash	−0.043 [−0.106, +0.007]
role	GPT-5 mini	−0.015 [−0.051, +0.022]
role	Qwen 3.8 Flash	−0.075 [−0.139, −0.016]
rendering	Mean over models	−0.094 [−0.123, −0.066]
rendering	Claude Haiku 4.5	−0.101 [−0.150, −0.054]
rendering	Gemini 3.7 Flash	−0.108 [−0.204, −0.043]
rendering	GPT-5 mini	−0.138 [−0.199, −0.090]
rendering	Qwen 3.8 Flash	−0.028 [−0.065, +0.011]
reference cases	Mean over models	−0.146 [−0.179, −0.117]
reference cases	Claude Haiku 4.5	−0.247 [−0.292, −0.204]
reference cases	Gemini 3.7 Flash	−0.111 [−0.174, −0.054]
reference cases	GPT-5 mini	−0.082 [−0.125, −0.038]
report	Mean over models	−0.204 [−0.247, −0.162]
report	Claude Haiku 4.5	−0.242 [−0.308, −0.181]
report	Gemini 3.7 Flash	−0.193 [−0.274, −0.125]
report	GPT-5 mini	−0.179 [−0.256, −0.115]
perceiver	Mean over models	−0.275 [−0.334, −0.218]
perceiver	Claude Haiku 4.5	−0.236 [−0.296, −0.179]
perceiver	Gemini 3.7 Flash	−0.303 [−0.373, −0.244]
perceiver	GPT-5 mini	−0.287 [−0.352, −0.225]
reasoner	Mean over models	−0.167 [−0.222, −0.129]
reasoner	Claude Haiku 4.5 and Gemini 3.7 Flash	−0.224 [−0.297, −0.166]
reasoner	Claude Haiku 4.5 and GPT-5 mini	−0.194 [−0.265, −0.144]
reasoner	Gemini 3.7 Flash and GPT-5 mini	−0.081 [−0.134, −0.043]
model	Mean over models	−0.473 [−0.544, −0.406]
model	Claude Haiku 4.5 and Gemini 3.7 Flash	−0.607

Table S22: What two readers share, MedXpertQA-MM: matched within-model contrasts of mean pairwise error correlation ($\Delta\phi$); sampling only, the baseline ϕ of generic-panel readers of one model.

Component	Model or model pair	$\Delta\phi$ (sampling only: ϕ)
sampling only	Claude Haiku 4.5	0.915 [0.885, 0.949]
sampling only	Gemini 3.7 Flash	0.811 [0.743, 0.863]
sampling only	GPT-5 mini	0.692 [0.642, 0.739]
sampling only	Qwen 3.8 Flash	0.629 [0.570, 0.685]
role	Mean over models	−0.033 [−0.051, −0.014]
role	Claude Haiku 4.5	−0.080 [−0.119, −0.041]
role	Gemini 3.7 Flash	−0.062 [−0.114, −0.018]
role	GPT-5 mini	−0.007 [−0.039, +0.022]
role	Qwen 3.8 Flash	+0.017 [−0.019, +0.054]
rendering	Mean over models	−0.045 [−0.062, −0.027]
rendering	Claude Haiku 4.5	−0.089 [−0.136, −0.048]
rendering	Gemini 3.7 Flash	−0.080 [−0.131, −0.033]
rendering	GPT-5 mini	−0.015 [−0.041, +0.009]
rendering	Qwen 3.8 Flash	+0.005 [−0.016, +0.028]
report	Mean over models	−0.172 [−0.211, −0.142]
report	Claude Haiku 4.5	−0.192 [−0.250, −0.144]
report	Gemini 3.7 Flash	−0.165 [−0.218, −0.110]
report	GPT-5 mini	−0.159 [−0.213, −0.104]
perceiver	Mean over models	−0.263 [−0.294, −0.227]
perceiver	Claude Haiku 4.5	−0.278 [−0.324, −0.227]
perceiver	Gemini 3.7 Flash	−0.249 [−0.292, −0.203]
perceiver	GPT-5 mini	−0.263 [−0.308, −0.218]
reasoner	Mean over models	−0.430 [−0.477, −0.389]
reasoner	Claude Haiku 4.5 and Gemini 3.7 Flash	−0.546 [−0.605, −0.488]
reasoner	Claude Haiku 4.5 and GPT-5 mini	−0.435 [−0.500, −0.379]
reasoner	Gemini 3.7 Flash and GPT-5 mini	−0.310 [−0.371, −0.256]
model	Mean over models	−0.492 [−0.536, −0.445]
model	Claude Haiku 4.5 and Gemini 3.7 Flash	−0.689 [−0.790, −0.614]
model	Claude Haiku 4.5 and GPT-5 mini	−0.483 [−0.567, −0.394]
model	Claude Haiku 4.5 and Qwen 3.8 Flash	−0.472 [−0.560, −0.398]
model	Gemini 3.7 Flash and GPT-5 mini	−0.493 [−0.580, −0.417]
model	Gemini 3.7 Flash and Qwen 3.8 Flash	−0.499

Table S23: Regression summary of pairwise error correlation (additive; one intercept per pair of reasoner models).

Set	Perceiver	Image vs report	Rendering	Role
VQA-RAD	−0.247 [−0.283, −0.210]	−0.123 [−0.173, −0.078]	−0.023 [−0.036, −0.009]	−0.006 [−0.016, +0.005]
SLAKE	−0.193 [−0.232, −0.146]	−0.094 [−0.135, −0.060]	−0.017 [−0.032, −0.003]	−0.004 [−0.015, +0.006]
MedXpertQA-MM	−0.160 [−0.187, −0.131]	−0.099 [−0.127, −0.071]	−0.012 [−0.021, −0.003]	−0.009 [−0.017, −0.002]

Table S24: Regression summary of pairwise error correlation (additive; one intercept per pair of reasoner models) (continued).

Set	Reference cases	Report	Reasoner (mean)
VQA-RAD	−0.026 [−0.043, −0.011]	+0.150 [+0.092, +0.218]	−0.026 [−0.045, −0.004]
SLAKE	−0.040 [−0.053, −0.028]	+0.009 [−0.051, +0.059]	−0.124 [−0.168, −0.086]
MedXpertQA-MM	+0.000 [+0.000, +0.000]	−0.066 [−0.104, −0.022]	−0.194 [−0.228, −0.164]

Table S25: Reference cases: accuracy (%) and error correlation ϕ with no references, shared references, and private references, with images or as text.

Set	Model	Single, none (%)	Single, shared images (%)	Single, shared text (%)	Panel, none (%)
VQA-RAD	GPT-5 mini	72.0 [66.4, 77.2]	73.6 [68.0, 78.8]	75.6 [70.0, 80.4]	73.6 [68.0, 78.8]
SLAKE	GPT-5 mini	80.8 [76.0, 85.6]	79.2 [74.4, 84.0]	81.6 [76.8, 86.4]	78.4 [73.2, 83.6]
VQA-RAD	Gemini 3.7 Flash	81.2 [76.4, 86.0]	82.4 [77.2, 87.2]	83.6 [78.8, 88.0]	80.4 [75.2, 84.8]
SLAKE	Gemini 3.7 Flash	91.2 [87.6, 94.4]	89.2 [85.2, 92.8]	90.8 [87.2, 94.4]	90.0 [86.4, 93.6]
VQA-RAD	Claude Haiku 4.5	60.8 [54.4, 66.8]	68.8 [63.2, 74.0]	61.2 [54.8, 67.2]	62.0 [55.6, 68.0]
SLAKE	Claude Haiku 4.5	72.4 [67.2, 78.0]	67.6 [62.0, 73.2]	76.8 [71.6, 82.0]	73.6 [68.0, 79.2]

Table S26: Reference cases (continued).

Set	Model	Panel, shared images (%)	Panel, private images (%)	Panel, shared text (%)	Panel, private text (%)
VQA-RAD	GPT-5 mini	74.8 [69.6, 80.4]	75.2 [69.6, 80.4]	76.4 [71.2, 81.6]	74.8 [69.6, 80.0]
SLAKE	GPT-5 mini	81.6 [76.8, 86.4]	84.0 [79.2, 88.4]	82.0 [77.2, 86.8]	81.6 [76.8, 86.4]
VQA-RAD	Gemini 3.7 Flash	82.8 [78.4, 87.2]	82.0 [77.2, 86.4]	84.4 [80.0, 88.8]	84.4 [79.6, 88.8]
SLAKE	Gemini 3.7 Flash	89.2 [85.6, 92.8]	90.4 [86.8, 93.6]	90.4 [86.4, 94.0]	92.0 [88.8, 95.2]
VQA-RAD	Claude Haiku 4.5	65.2 [58.8, 71.2]	64.4 [58.8, 70.0]	62.4 [56.0, 68.4]	62.4 [56.0, 68.0]
SLAKE	Claude Haiku 4.5	70.4 [64.8, 76.0]	68.8 [62.8, 74.8]	78.0 [73.2, 83.2]	78.0 [73.2, 82.8]

Table S27: Reference cases (continued).

Set	Model	ϕ none	ϕ shared images	ϕ private images	ϕ shared text
VQA-RAD	GPT-5 mini	0.741 [0.683, 0.797]	0.751 [0.694, 0.799]	0.668 [0.608, 0.723]	0.737 [0.681, 0.792]
SLAKE	GPT-5 mini	0.800 [0.746, 0.844]	0.741 [0.667, 0.807]	0.667 [0.600, 0.737]	0.780 [0.718, 0.842]
VQA-RAD	Gemini 3.7 Flash	0.776 [0.715, 0.836]	0.708 [0.640, 0.772]	0.673 [0.599, 0.744]	0.743 [0.658, 0.816]
SLAKE	Gemini 3.7 Flash	0.828 [0.744, 0.895]	0.836 [0.761, 0.894]	0.722 [0.629, 0.797]	0.823 [0.725, 0.891]
VQA-RAD	Claude Haiku 4.5	0.878 [0.835, 0.911]	0.890 [0.849, 0.928]	0.664 [0.612, 0.713]	0.912 [0.878, 0.942]
SLAKE	Claude Haiku 4.5	0.882 [0.838, 0.922]	0.928 [0.894, 0.955]	0.655 [0.608, 0.708]	0.941 [0.907, 0.970]

Table S28: Reference cases (continued).

Set	Model	ϕ private text
VQA-RAD	GPT-5 mini	0.703 [0.637, 0.759]
SLAKE	GPT-5 mini	0.690 [0.620, 0.752]
VQA-RAD	Gemini 3.7 Flash	0.692 [0.618, 0.763]
SLAKE	Gemini 3.7 Flash	0.716 [0.599, 0.804]
VQA-RAD	Claude Haiku 4.5	0.681 [0.630, 0.739]
SLAKE	Claude Haiku 4.5	0.720 [0.657, 0.779]

Table S29: Open-weight models: accuracy (%) and visual reliance (|change in log-odds of yes| on yes/no items; KL divergence on MedXpertQA-MM).

Model	Set	Accuracy image (%)	Accuracy gray (%)	256 px	128 px
HuatuoGPT-Vision 7B	VQA-RAD	74.8 [69.6, 80.0]	56.4 [50.4, 62.4]	77.2 [72.0, 82.4]	75.6 [70.4, 80.8]
HuatuoGPT-Vision 7B	SLAKE	76.8 [72.0, 82.0]	50.8 [44.4, 57.2]	76.0 [70.8, 81.2]	76.0 [70.8, 81.6]
HuatuoGPT-Vision 7B	MedXpertQA-MM	20.0 [15.2, 25.2]	20.0 [15.2, 25.2]	19.6 [15.2, 24.4]	20.4 [15.6, 25.2]
LLaVA-1.5 7B	VQA-RAD	56.0 [49.6, 62.0]	54.0 [48.0, 60.0]	56.4 [50.4, 62.4]	56.0 [49.6, 62.0]
LLaVA-1.5 7B	SLAKE	66.0 [59.6, 72.0]	50.0 [43.6, 56.4]	66.4 [60.4, 72.4]	66.0 [60.4, 72.0]
LLaVA-1.5 7B	MedXpertQA-MM	21.2 [16.4, 26.4]	20.4 [15.6, 25.6]	21.2 [16.4, 26.4]	20.8 [16.0, 25.6]
MedGemma 4B	VQA-RAD	80.0 [75.2, 84.8]	60.8 [54.8, 66.8]	80.4 [75.6, 85.2]	76.0 [71.2, 81.2]
MedGemma 4B	SLAKE	88.8 [85.2, 92.4]	40.0 [34.0, 46.4]	86.8 [82.8, 91.2]	84.0 [79.6, 88.4]
MedGemma 4B	MedXpertQA-MM	22.0 [16.8, 27.2]	23.2 [18.0, 28.4]	22.4 [17.2, 27.6]	22.4 [17.2, 27.6]
Qwen2-VL 7B	VQA-RAD	76.8 [71.6, 82.0]	58.8 [53.2, 64.4]	73.6 [68.4, 78.8]	76.4 [70.8, 81.6]
Qwen2-VL 7B	SLAKE	78.0 [73.2, 83.2]	43.2 [36.8, 49.6]	72.8 [67.6, 78.4]	68.0 [62.4, 74.0]
Qwen2-VL 7B	MedXpertQA-MM	20.8 [16.0, 26.0]	21.6 [16.8, 26.8]	21.2 [16.4, 26.4]	21.2 [16.4, 26.4]
Qwen3-VL 8B	VQA-RAD	74.0 [68.4, 79.2]	57.6 [51.6, 63.6]	70.0 [64.0, 75.6]	68.8 [62.8, 74.4]
Qwen3-VL 8B	SLAKE	79.2 [74.0, 84.4]	43.6 [37.6, 49.6]	83.6 [78.8, 88.0]	81.2 [76.4, 86.0]
Qwen3-VL 8B	MedXpertQA-MM	24.4 [19.2, 30.0]	23.6 [18.8, 28.8]	25.6 [20.4, 31.2]	25.2 [20.0, 30.4]

Table S30: Open-weight models (continued).

Model	Set	Visual reliance, mean	median
HuatuoGPT-Vision 7B	VQA-RAD	0.89 [0.77, 1.01]	0.62
HuatuoGPT-Vision 7B	SLAKE	1.84 [1.66, 2.03]	1.50
HuatuoGPT-Vision 7B	MedXpertQA-MM	0.10 [0.07, 0.13]	0.04
LLaVA-1.5 7B	VQA-RAD	1.10 [1.02, 1.18]	1.12
LLaVA-1.5 7B	SLAKE	1.40 [1.29, 1.52]	1.25
LLaVA-1.5 7B	MedXpertQA-MM	0.10 [0.07, 0.14]	0.02
MedGemma 4B	VQA-RAD	6.43 [5.71, 7.17]	4.38
MedGemma 4B	SLAKE	11.23 [10.58, 11.95]	11.63
MedGemma 4B	MedXpertQA-MM	0.88 [0.65, 1.13]	0.02
Qwen2-VL 7B	VQA-RAD	1.49 [1.33, 1.64]	1.12
Qwen2-VL 7B	SLAKE	2.44 [2.21, 2.66]	2.13
Qwen2-VL 7B	MedXpertQA-MM	0.11 [0.08, 0.14]	0.04
Qwen3-VL 8B	VQA-RAD	5.98 [5.36, 6.61]	4.38
Qwen3-VL 8B	SLAKE	9.72 [8.95, 10.55]	9.07
Qwen3-VL 8B	MedXpertQA-MM	0.72 [0.55, 0.93]	0.07

Table S31: Open-weight models: error correlation between two models.

Set	Model 1	Model 2	ϕ	$\phi/\phi_{\max}$	Shared encoder
VQA-RAD	HuatuoGPT-Vision 7B	LLaVA-1.5 7B	0.172	0.263	yes
VQA-RAD	HuatuoGPT-Vision 7B	MedGemma 4B	0.332	0.385	no
VQA-RAD	HuatuoGPT-Vision 7B	Qwen2-VL 7B	0.227	0.239	no
VQA-RAD	HuatuoGPT-Vision 7B	Qwen3-VL 8B	0.412	0.421	no
VQA-RAD	LLaVA-1.5 7B	MedGemma 4B	0.181	0.321	no
VQA-RAD	LLaVA-1.5 7B	Qwen2-VL 7B	0.162	0.261	no
VQA-RAD	LLaVA-1.5 7B	Qwen3-VL 8B	0.099	0.148	no
VQA-RAD	MedGemma 4B	Qwen2-VL 7B	0.294	0.323	no
VQA-RAD	MedGemma 4B	Qwen3-VL 8B	0.365	0.432	no
VQA-RAD	Qwen2-VL 7B	Qwen3-VL 8B	0.322	0.348	no
SLAKE	HuatuoGPT-Vision 7B	LLaVA-1.5 7B	0.346	0.451	yes
SLAKE	HuatuoGPT-Vision 7B	MedGemma 4B	0.316	0.488	no
SLAKE	HuatuoGPT-Vision 7B	Qwen2-VL 7B	0.371	0.385	no
SLAKE	HuatuoGPT-Vision 7B	Qwen3-VL 8B	0.535	0.574	no
SLAKE	LLaVA-1.5 7B	MedGemma 4B	0.227	0.459	no
SLAKE	LLaVA-1.5 7B	Qwen2-VL 7B	0.373	0.504	no
SLAKE	LLaVA-1.5 7B	Qwen3-VL 8B	0.360	0.505	no
SLAKE	MedGemma 4B	Qwen2-VL 7B	0.485	0.725	no
SLAKE	MedGemma 4B	Qwen3-VL 8B	0.224	0.324	no
SLAKE	Qwen2-VL 7B	Qwen3-VL 8B	0.418	0.433	no
MedXpertQA-MM	HuatuoGPT-Vision 7B	LLaVA-1.5 7B	0.205	0.213	yes
MedXpertQA-MM	HuatuoGPT-Vision 7B	MedGemma 4B	0.338	0.359	no
MedXpertQA-MM	HuatuoGPT-Vision 7B	Qwen2-VL 7B	0.507	0.520	no
MedXpertQA-MM	HuatuoGPT-Vision 7B	Qwen3-VL 8B	0.275	0.312	no
MedXpertQA-MM	LLaVA-1.5 7B	MedGemma 4B	0.244	0.250	no
MedXpertQA-MM	LLaVA-1.5 7B	Qwen2-VL 7B	0.192	0.195	no
MedXpertQA-MM	LLaVA-1.5 7B	Qwen3-VL 8B	0.161	0.176	no
MedXpertQA-MM	MedGemma 4B	Qwen2-VL 7B	0.227	0.236	no
MedXpertQA-MM	MedGemma 4B	Qwen3-VL 8B	0.238	0.254	no
MedXpertQA-MM	Qwen2-VL 7B	Qwen3-VL 8B	0.282	0.313	no